China and Africa: A Quantitative Analysis of the Political Relationship in the Hu Jintao and Xi Jinping Eras

Introduction

Research on the political relationship between China and Africa is an integral part of the scholarly literature on China. This literature focuses on events and connections spanning thousands of years, from 100 BCE to the contemporary age. Looking at the distant past, for example, is Gao's (1984) paper, which discusses Si Machien's ancient historical treatise, the Shi Ji, and its mention of a mysterious location that an explorer of Emperor Wuti's (Han Dyansty) visited—Likan. He goes on to discuss historian's debates over whether Likan is truly, as many argue, Alexandria in Egypt, making this explorer's visit the earliest contact of China with Africa. Coming closer to the present, Ismael (1971) looks at cooperation between the newly-minted People's Republic of China (PRC) and countries in Africa after the Bandung conference of 1955, with an eye to the communist rhetoric surrounding this phase of the relationship. Taylor (1998) jumps forward to the 1990s, exploring PRC-Africa relations following years of Deng Xiaoping's major economic reforms, while van Staden (2018) brings readers a few years off of the present day, focusing on the 19th Communist Party Congress (held in 2017) and its signals on a new turn in Sino-African relations.

However, despite the vastness of China-Africa political research, of which only a few key articles are mentioned above, a key issue remains in the field: the lack of quantitative analysis. All the articles discussed earlier, though they are major cornerstones of this field, approach China-Africa political relations from a qualitative perspective, and they tend to lean on traditional historiographical and political analysis methods. For example, Ismael's (1971) paper goes into extreme detail on conferences held, visits done, and treaties signed between the PRC

and various African nations in the 1950s and 60s, but he never uses any basic document analysis to back up his claims of growing closeness between the PRC and Africa in these two decades. To be sure, quantitative analysis was in its infancy in the 1970s, when Ismael wrote "The People's Republic of China and Africa," but the absence of quantitative analysis plagues China-Africa research after the computer revolution of the 1990s, as Taylor's (1998) paper exemplifies. Taylor does bring in figures to his research, but these are purely economic ones—trade and foreign aid between the two regions in the 1990s—with the brunt of his argument resting on an authoritative telling of the historical narrative of China and Africa in this period.

The absence of quantitative analysis is especially glaring in research focused on the (near) present. With the rise of Xi Jinping and his new, more aggressive foreign policy perspective to the highest leadership position in the PRC, more and more China scholars are focusing on the new approach Xi brings to the PRC and its slow rise in the country. Scobell and Harold (2013), writing shortly after Xi's election to CCP (Chinese Communist Party) General Secretary and President of the PRC, use interviews with Shanghai analysts to argue that the PRC's increasingly assertive stance in world politics between 2008 and 2010 was a result of perceived increases in US aggression towards the PRC. Yan (2014) takes the theory of moral realism to analyze Xi's new "striving for achievement" rhetoric in this realm and compare it to Deng Xiaoping's completely opposing rhetoric, "keeping a low profile." And yet, despite great interest, large corpuses (with most PRC policy papers and speeches easily found online, as well as social media posts from government representatives), and a vast increase in the computing capacity of a basic computer, only one paper so far has taken the opportunity to use quantitative methods to analyze shifts in the PRC's foreign policy rhetoric since Xi Jinping's rise—Dai and Luqiu's (2022) analysis of "wolf warrior" rhetoric using a large corpus of their speeches.

As a result, this article aims to contribute to both the literature on contemporary China-Africa relations and Xi Jinping's new foreign policy rhetoric through the use of quantitative methods, with the hope of showcasing the power these methods hold for both fields. It uses bigrams and trigrams, correspondence analysis, Word2Vec, and t-SNE to analyze an innovative corpus of the questions asked and answers given at the PRC's Ministry of Foreign Affairs daily press conferences, comparing questions and answers related to Africa across the Hu Jintao and Xi Jinping years. I find that Xi Jinping indeed has more intense rhetoric than Hu Jintao in the foreign policy sphere, but he is not as fixated on Africa as qualitative analysis may suggest; furthermore, an additional dimension, China-focused vs. world-focused, can be used to describe years across and within the Hu and Xi eras, and Hu and Xi's similar Africa partnership rhetoric has slowly moved from state speech to widespread acceptance among journalists.

A Brief Overview of China-Africa Relations

Historical Relations

Before 1949, the year the CCP won its civil war against the KMT/GMD (Kuomintang/Guomindang, depending on romanization), China already had a long history of engaging with Africa, though indirectly. As Gao (1984) explains (and was mentioned earlier), some historians believe that China's earliest contact with the African continent was under the Han Dynasty's Emperor Wuti in 100 BCE, when this emperor sent an envoy to the western lands who reached the area of Likan (thought to be Alexandria, Egypt) (241). Even if one contends this interpretation of Likan, other facts point to, at the very least, Chinese goods reaching north Africa shortly before and during the Common Era (and vice-versa). Gao (1984) references stories describing Queen Cleopatra (reign 51 BCE to 30 BCE) wearing Chinese silk, as well as

Emperor Wenti receiving elephant tusks and rhinoceros horns in 166 CE from Roman Emperor Antonias (242).

From this early period, indirect contact between Africa and China continued through the Common Era dynasties and slowly grew into direct contact. African and Chinese goods were traded, mostly through Arab merchants, during the Tang (618 CE - 907 CE) Sung (960 CE - 1279 CE), and Yuan (1279 CE - 1368 CE) dynasties, a relationship that finally culminated in reciprocated visits during the Ming dynasty (1368 CE - 1644 CE), when Admiral Zhang He sailed to the eastern coast of Africa and made political connections with the local kings in modern-day Somalia and Kenya (Alden and Alves 2008). However, these connections were quickly shut down by the later Ming isolationist policy, which continued during the Qing dynasty (1644 CE - 1911 CE), and China-Africa relations found themselves at an all-time low under the Guomindang government in the interwar period, which was too consumed with internal strife and the Japanese threat to form connections far abroad (Alden and Alves 2008).

Under communist rule, China, now the People's Republic of China, began to draw closer to Africa. Its first major overture to countries on the continent was its "Five Principles of Peaceful Coexistence," proclaimed shortly after the communist victory in 1949, which was meant to entice newly-independent nations with its non-interference and non-aggression rhetoric (Alden and Alves 2008). A few years later, Zhou Enlai, the PRC foreign minister, attended the Bandung Conference of 1955, where he met President Nasser of Egypt and forged connections with other African leaders, resulting in Egypt recognizing the PRC in May 1956 (Ismael 1971, 507). PRC engagement with Africa continued to grow as more and more nations gained independence in the late 1950s and early 1960s, with Zhou Enlai and Chen Yi, his successor, making a tour of Africa in 1964-1965 and gaining the recognition of 15 more African countries

(Yu 1977, 97). The 1960s, though, ended on a low point for China-Africa relations, with warming gradually occurring in the 1970s (Yu 1977, 98).

Under Deng Xiaoping, who came into power a few years after the death of Mao Zedong in 1976, the PRC didn't intentionally distance itself from Africa, but Deng's focus on the domestic PRC economy and the country's modernization meant that he focused little on Africa, a continent that could do little to aid him in achieving his goals (Taylor 1998, 444). It wasn't until the stunning Western condemnation of the Tiananmen Square massacre on June 4, 1989, that Deng began to once again focus on Africa, realizing that his positive relationship with the West was taking a sour tone and he would need new allies (Taylor 1998, 446). He and his successor, Jiang Zeming, increased military, political, and economic ties with African countries, showcased at the highest level by President Yang Shangkun and Jiang Zeming's visits to Africa in 1992 and 1995, respectively, as well as the billions of dollars in weapons shipped to Ethiopia, the Congo, and Sierra Leone (Muekalia 2004, 8). Trade between Africa and China also shot up during this period—$10 billion USD in 2000 (Muekalia 2004, 8).

Relations under Hu Jintao

Scholars typically characterize China-Africa relations under Hu Jintao, who became president on March 15th, 2003, as building on the foundation set by Deng Xiaoping and Jiang Zeming. Under Hu Jintao, PRC Foreign Direct Investment (FDI) in Africa skyrocketed, reaching $8 billion in 2006 and complemented by the Forum on China-Africa Cooperation (FOCAC), which started in 2006 and created a $5 billion China-Africa Development Fund (Sautman and Yang 2007, 79). The PRC heavily invested in infrastructure, especially for the extraction of natural resources, offering low-interest or zero-interest loans that could be repaid with these same natural resources (Sautman and Yang 2007, 80), looking specifically for oil imports (Wang

and Zou 2014, 1114). The PRC also began to increase its so-called "soft power" in Africa by expanding beyond pure economic relations into other development spheres. For example, at the fifth conference of FOCAC, Hu Jintao announced the PRC's intention to help train 30,000 African professionals, as well as provide students with 18,000 government scholarships (Wang and Zou 2014, 1117). Finally, during this period, the PRC began to get more involved in conflicts and conflict-management in Africa, notably convincing the Khartoum government in 2007 to allow a joint United Nations-African Union peacekeeping force in Darfur and showing concern over the NATO intervention in Libya in 2011 (Mariani 2002, 28). Thus, I expect Sudan- and Libya- related words to show up in the corpus under analysis for the Hu Jintao years, as well as FOCAC-related words and words related to natural resources and infrastructure.

Relations under Xi Jinping

With Xi Jinping coming to power in 2013, he continued to engage in Africa, seeming to push PRC-Africa relations to new highs. The PRC sent its first ever security troops on a UN peacekeeping mission in June 2013, to Mali, a stunning shift from its no-combat-troop-contribution policy (Wang and Zou 2014, 1120). In 2016, three years after Xi's ascent, PRC investments in Africa broke $100 billion, driven in part by the Xi's flagship program, the Belt and Road Initiative (BRI), which continues to be critical to heightening PRC engagement with Africa (van Staden 2018, 2). 37 out of 58 (around 64%) African countries and the African Union are officially part of the BRI, with major projects including the Addis Ababa-Djibouti Railway in Ethiopia and Djibouti ($1.3 billion borrowed), the Karuma hydropower plant in Uganda ($1.4 billion borrowed), and the Abuja-Masaka light rail in Nigeria ($500 million borrowed) (Dollar 2019, 2-4). Thus, I expect that the Xi Jinping Africa corpus will contain words related to these

projects, the BRI, as well as more words related to UN peacekeeping missions in comparison to the Hu Jintao years, with investment-related words remaining constant or increasing slightly.

Methods

Corpus

To quantitatively analyze the relationship between Africa and China during the Hu Jintao and Xi Jinping years and identify differences between the two, I used an innovative new corpus from Mochtak and Turcsanyi (2021). They webscraped the PRC Ministry of Foreign Affairs (MFA) page to collect all the daily press conference transcripts from 2002 up to 2023 and divided up the transcripts into question-answer pairs, creating a corpus that includes the date of the question-answer, the spokesperson who answered the question, and locations mentioned in the question and the answer in separate columns. This corpus is particularly useful for the aims of this project because, as Mochtak and Turcsanyi (2021) themselves explain, the MFA curates the transcripts before publishing them online, meaning that they take out any unfriendly questions, correct poor wording on the part of the spokesperson, and change other parts of the transcript to present the state's preferred narrative. As a result, this corpus uniquely showcases the exact signals the PRC wants to send to the international community, allowing for easy translation into comparisons of the Hu Jintao and Xi Jinping rhetoric on Africa.

Data Cleaning

As the MFA corpus was already divided into question-answer pairs and location tags existed for both the question and the answer, all that was left was to tag questions and answers as relating to Africa or not. For this, I used the a-world-of-countries database to not only screen for the word "africa" in the question and answer location lists, but also African countries and capitals. This resulted in about 2,944 locations needing to be checked manually, which was done

with the aid of the internet and my own geographical knowledge. Some question locations and answer locations were left with only a "-", which I left as is, because there was no clear pattern of what the dash meant, as the question/answer definitely had locations within it.

Data Analysis

To analyze the data, I first chose to analyze questions and answers separately because Africa-related questions occasionally did not have Africa-related answers (as in the case of passing mentions) and vice-versa. I started out by generating counts of Africa-related questions and answers by year for descriptive analysis, and then I moved on to generating counts of statistically significant bigrams and trigrams in each year's questions and answers using the spacy and nltk packages, as well as sklearn's CountVectorizer to perform correspondence analysis. I then plotted the screeplot for both the question and answer dataframes as well as the questions and answers and years for both the questions and answers on two dimensions. Next, I applied Word2Vec using the Skipgram model to corpuses divided into Hu Jintao and Xi Jinping eras, using t-SNE to reduce all the vectors and the top 50 in each corpus for visualization. Results are below.

Results

Descriptive Analysis: Africa-Related Questions over The Years

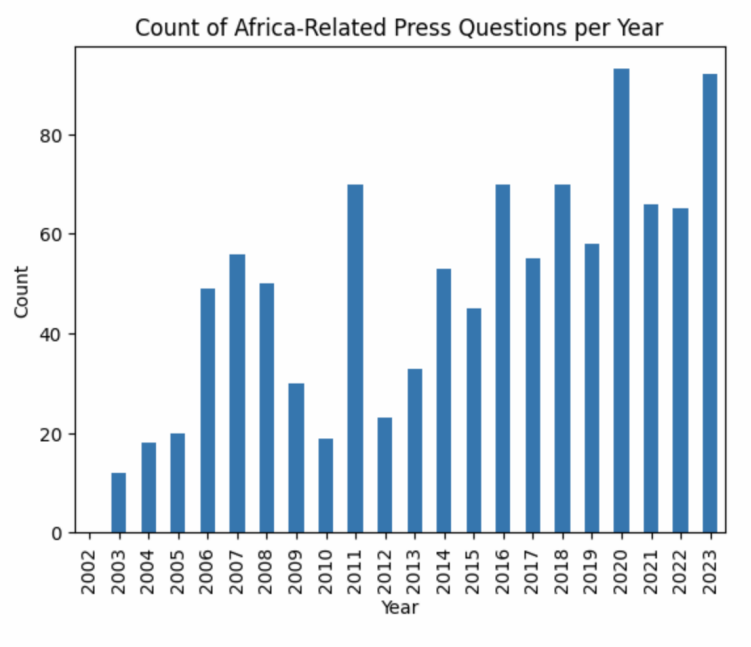


Figure 1. Count of Africa-Related Press Questions by Year.

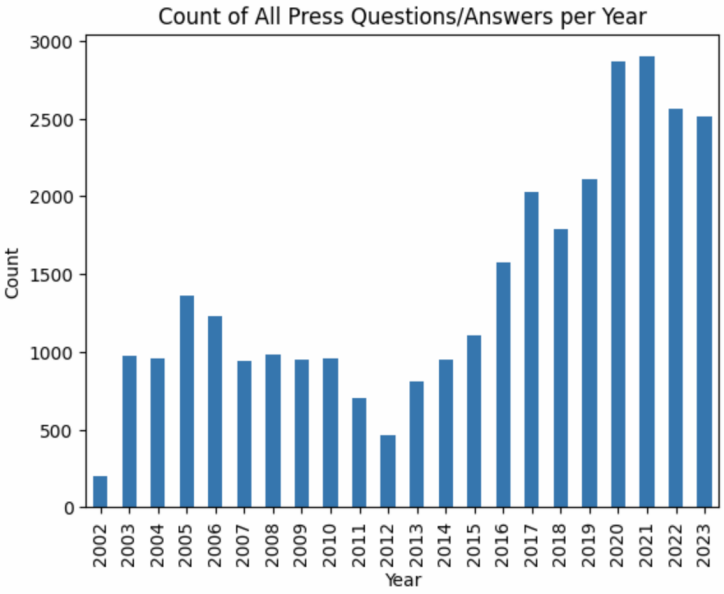


Figure 2. Count of All Press Questions/Answers by Year. As the corpus comes in question-answer pairs, the number of questions is the same as the number of answers.

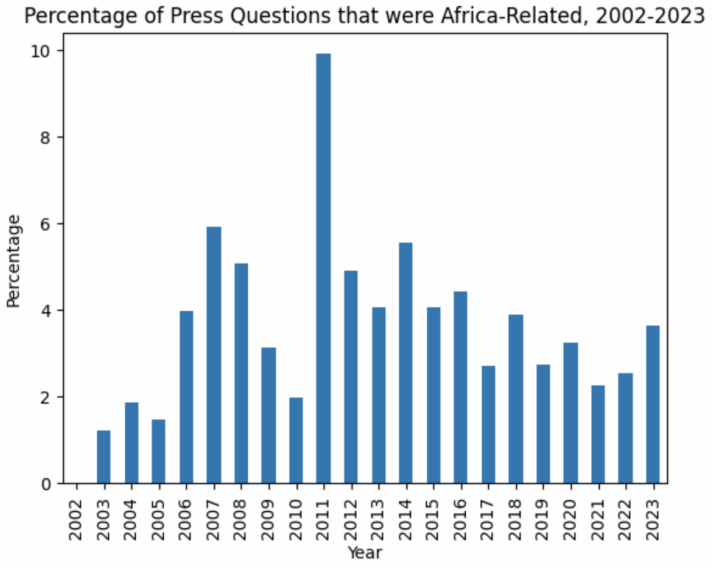


Figure 3. Percentage of Press Questions that were Africa-Related by Year.

As shown in Figure 1, Africa-related questions did not reach above 100 in any given year, with the highest peaks being 93 and 92 (2020 and 2023, respectively) and the lowest being 0 (2002). Local peaks happened around 2007 and 2011 (which was a strong outlier in comparison to the years around it), with question numbers slowly climbing from 2012. In contrast, as shown in Figure 2, press questions overall reached over 2,500 at their peaks in 2020 and 2021 (2869 and 2896, respectively) and were similarly at their lowest in 2002 and 2012 (200 and 467, respectively). As a result, the proportion of press questions that were Africa-related was at its highest in 2011 (9.92%), corresponding to a peak in Africa-related questions and an overall dip in all press questions, while the proportional low was 2002, when, because no Africa-related questions were asked, the figure was 0% (Figure 3). The proportional graph (Figure 3) shows local maximums in 2007 (5.93%) and a slow, creeping decline in the proportion of Africa-related questions asked from 2012 to 2023, though the values fluctuate around 3%.

Descriptive Analysis: Africa-Related Answers over The Years

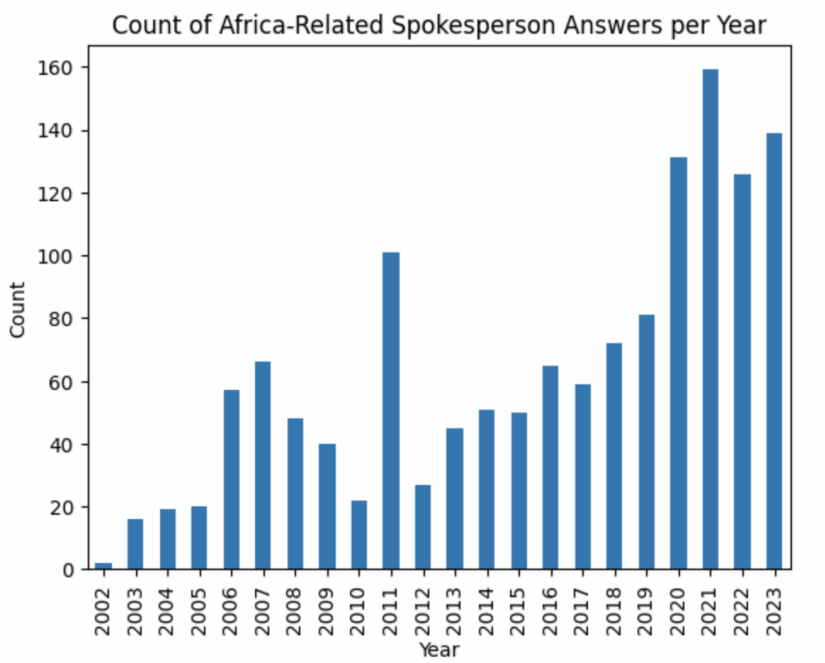


Figure 4. Count of Africa-Related MFA Spokesperson Answers by Year.

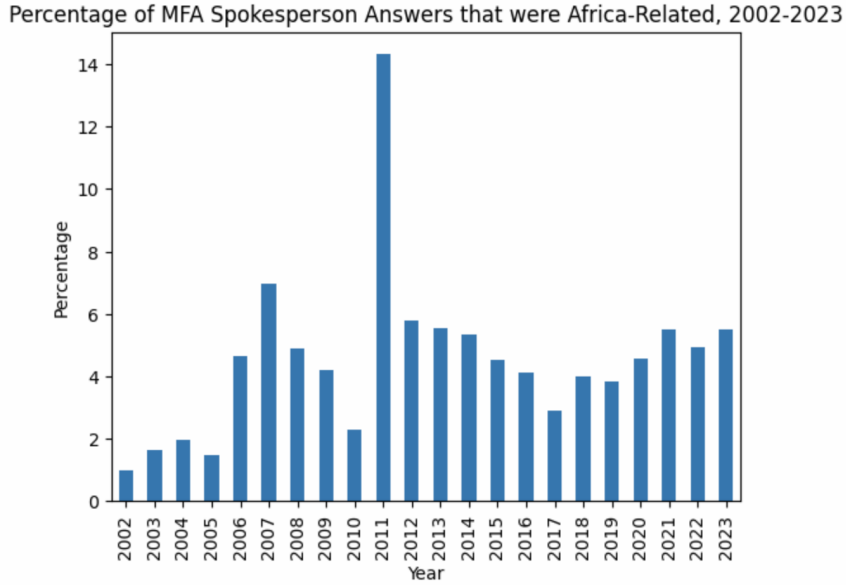


Figure 5. Percentage of MFA Spokesperson Answers related to Africa by Year.

The number of Africa-related MFA spokesperson answers over the years (Figure 4) showcases similar trends to the number of Africa-related questions over the years (Figure 1). Like the Africa-related questions, Africa-related answers have a peak in 2011, though this is of 101, higher than the number of Africa-related questions in that year by 31. Africa-related

answers also begin to climb after a low in 2012, though this growth is much less variable than that of the Africa-related questions. Africa-related answers also have higher peaks than Africa-related questions, reaching a peak of 159 in 2021 in comparison to the Africa-related questions peak of 93 in 2020. In terms of percentages (Figure 5), the percentage of MFA spokesperson answers that were Africa-related shows a much more distinct high in 2011, with Africa-related answers making up 14.31% of press answers in that year. Outside of that peak, the percentage of answers that are Africa-related seems to vary around 3%, similarly to the Africa-related questions percentage, though with less apparent variation and also a local peak in 2007.

Correspondence Analysis

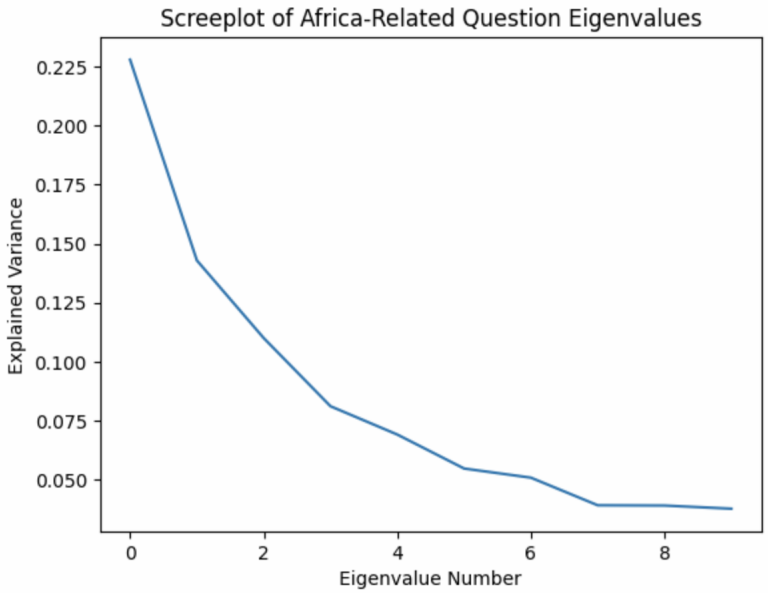


Figure 6. Screeplot of Africa-Related Question Eigenvalues. This screeplot shows the eigenvalues following correspondence analysis of the Africa-related questions corpus on the x-axis in descending order of explained variance, with the variance each dimension explains on its own on the y-axis.

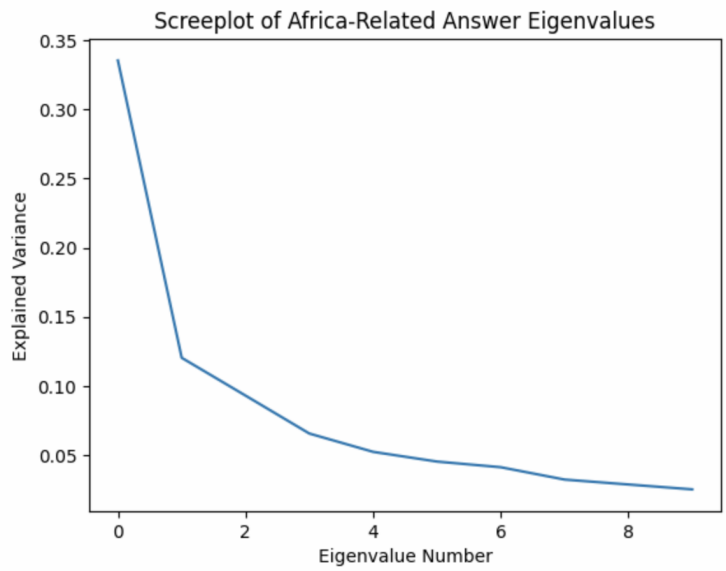


Figure 7. Screeplot of Africa-Related Answer Eigenvalues. This screeplot shows the eigenvalues following correspondence analysis of the Africa-related answers corpus on the x-axis in descending order of explained variance, with the variance each dimension explains on its own on the y-axis.

Both the screeplot of the Africa-related question (Figure 6) and Africa-related answer (Figure 7) corpus have a sharp elbow at the second dimension, suggesting that 2 dimensions is enough to explain much of the variance in both corpuses. The variance explained by these two dimensions, however, varies between the Africa-related question and Africa-related answer corpuses. For the former, around 22.5% of the variance in the original corpus is explained by the first dimension, as opposed to about 33% for the latter, and about 14% of the variance in the original corpus is explained by the second dimension for the question corpus, as opposed to about 12% for the latter. As a result, both dimensions together explain about 36.5% of the variance in the questions corpus and 45% in the answer corpus.

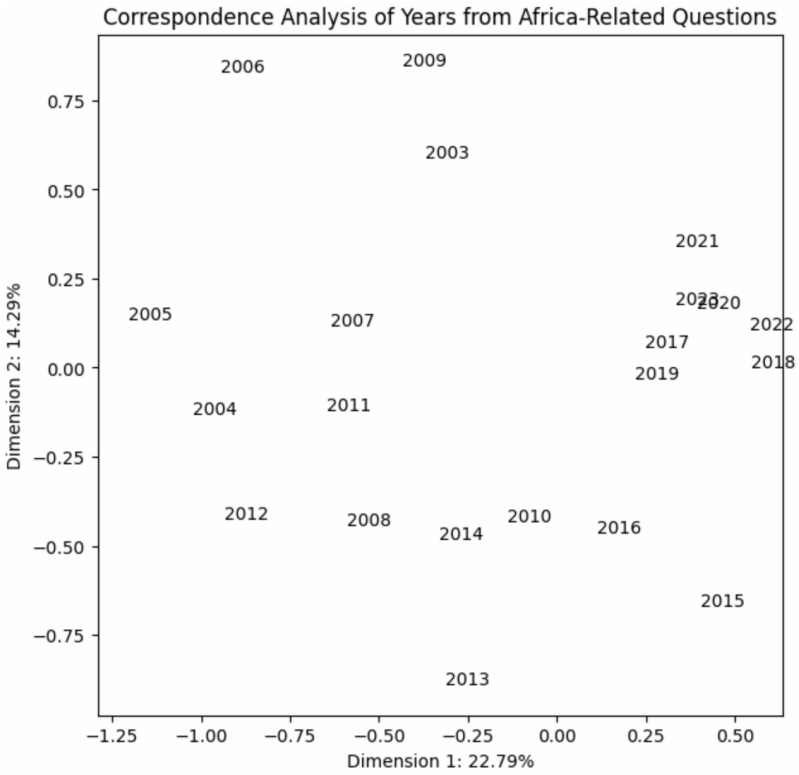


Figure 8. Graphical Representation of Correspondence Analysis on Africa-Related Press Questions, Years. Correspondence analysis was run on bigram/trigrams from the MFA press questions corpus by year.

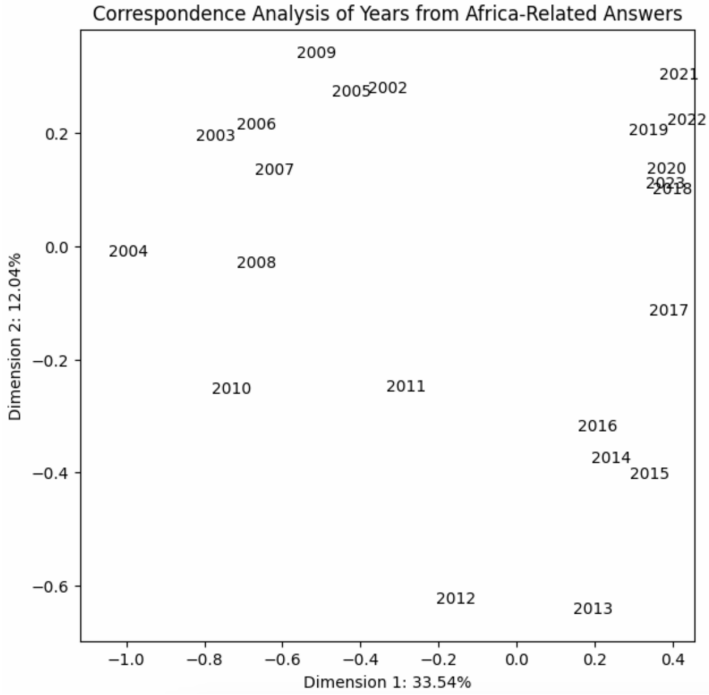


Figure 9. Graphical Representation of Correspondence Analysis on Africa-Related MFA Spokesperson Answers, Years. Correspondence analysis was run on bigram/trigrams from the MFA answer corpus by year.

Per Figure 8, for Africa-related questions, along dimension 1, which explains 22.79% of the original variation in the corpus, the extreme left is taken up by 2005 and the extreme right by a cluster of years including 2017, 2018, 2019, 2021, 2021, 2022, and 2023. On the other hand, at the bottom (along dimension 2, explaining 14.29% of variance in the Africa-related questions corpus) is the year 2013, while at the top are 2006 and 2009. Per Figure 9, the pattern is similar for Africa-related answers, though the second dimension is skewed upwards. 2004 is on the far left of dimension 1, explaining 33.54% of the variance in the Africa-related answers dataset, while 2018, 2019, 2020, 2021, 2022, and 2023 are all clustered on the far right, similarly to in the Africa-related questions correspondence analysis visualization. On the bottom, along dimension 2, which explains 12.04% of the original variance in the Africa-related answers data, lie 2012 and 2013, while at the far top is 2009, with 2005, 2002, and 2021 close below it.

Word2Vec and t-SNE

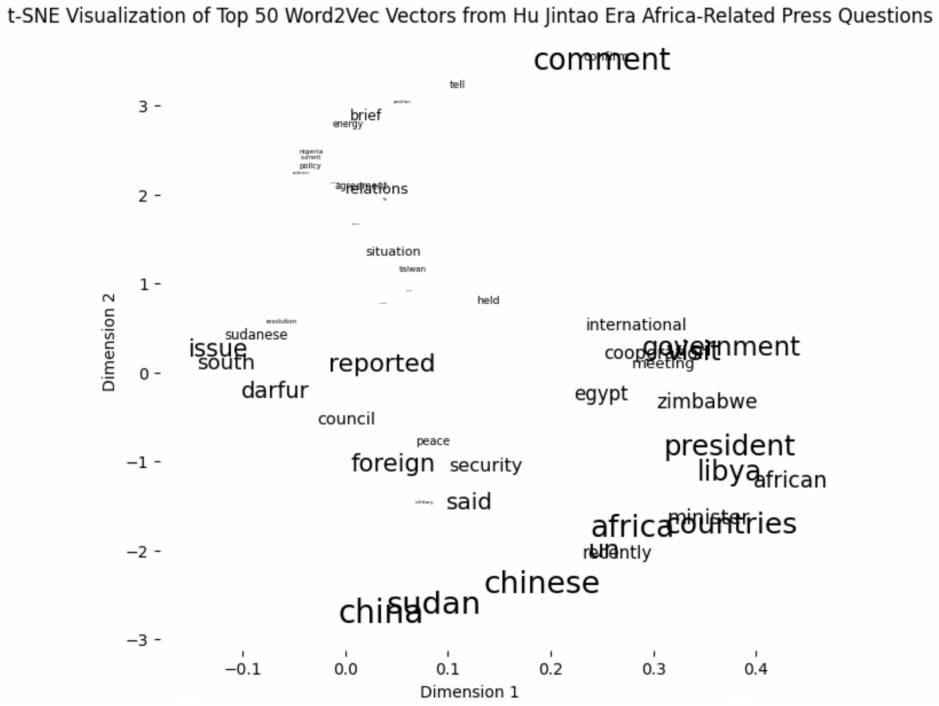


Figure 10. Visualization of Top 50 Word2Vec Vectors from the Hu Jintao Era Africa-Related Press Questions. Visualization on two dimensions using t-SNE reduction of the Word2Vec vectors. The larger the word, the more frequent it is in the corpus.

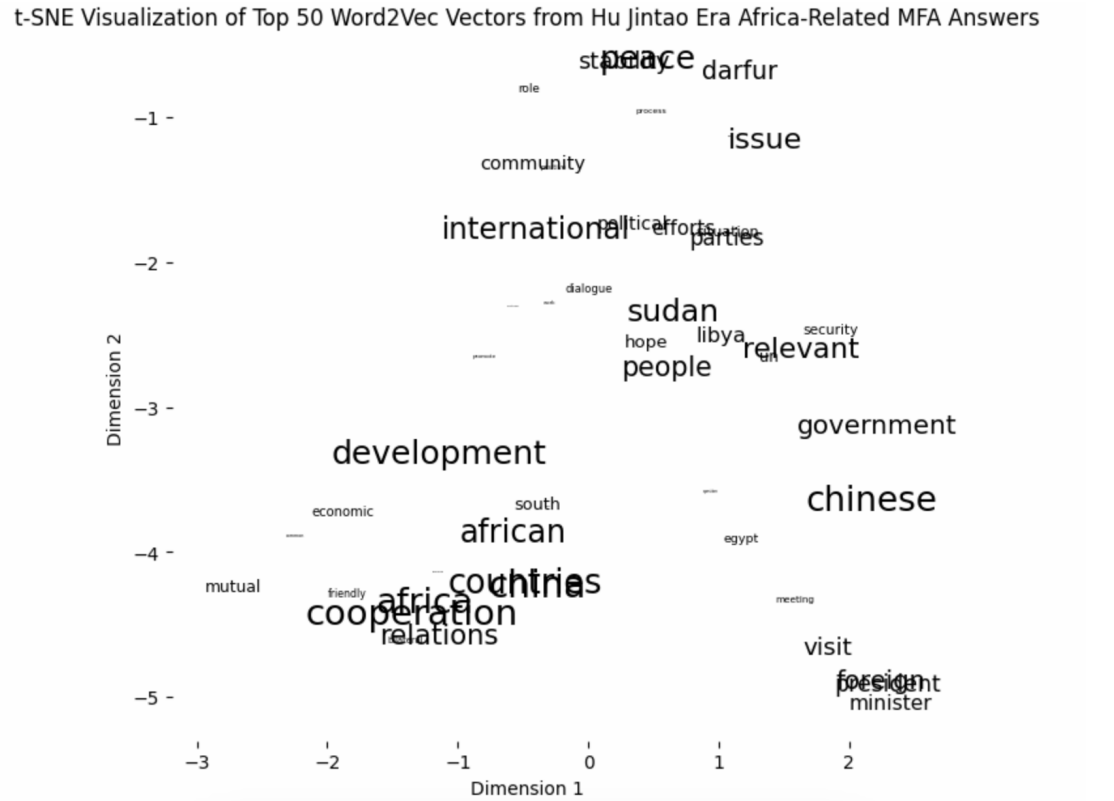


Figure 11. Visualization of Top 50 Word2Vec Vectors from the Hu Jintao Era Africa-Related MFA Answers. Visualization on two dimensions using t-SNE reduction of the Word2Vec vectors. The larger the word, the more frequent it is in the corpus.

As shown in Figure 10, on the far left of dimension 1 for Hu Jintao Era Africa-related press questions is the word issue, along with south, Sudanese, and Drfur, while the far right on this dimension is populated by government, cooperation, meeting, and international. The bottom (along dimension 2) has China, Sudan, and Chinese, while the top holds comment. By the size of the plotted words, the higher frequency words in this corpus are comment, countries, government, China, Sudan, Chinese, Africa, president, and Libya while the lower frequency words out of the top 50 are summit, development, and assistance. In contrast, as shown in Figure 11, on the far left of the first dimension for Hu Jintao Era Africa-related MFA answers lies the word mutual, while the far right is taken up by government, Chinese, foreign, president, and

minister. On the bottom (along dimension 2), lie foreign, minister, and president, and on the top lie peace and Darfur. The most frequent words here are similar to those for the Africa-related questions corpus: China, cooperation, Africa, and Chinese, while the least frequent words are developing, sides, and early.

For the Xi Jinping press questions corpus (Figure 11, below), the far left on the first dimension is taken up by election and national, while the far right is taken up by comment. The bottom (along the second dimension) is taken up by several words, including visit and foreign, while the far top has diplomatic. The most frequent words here are China, comment, Africa, and Chinese, while the least frequent of the top 50 are reported, economic, Kenya, and general. For the Xi Jinping MFA answer corpus, the left is taken up by Sudan, south, and relevant, while the right holds president and Xi. The top includes foreign, political, and minister, while the bottom has shared and rights. The top words are similar to those in the question corpus—China, countries, Africa, cooperation, while the lowest frequency holders of the top 50 are Arab, Covid-19, good, and council.

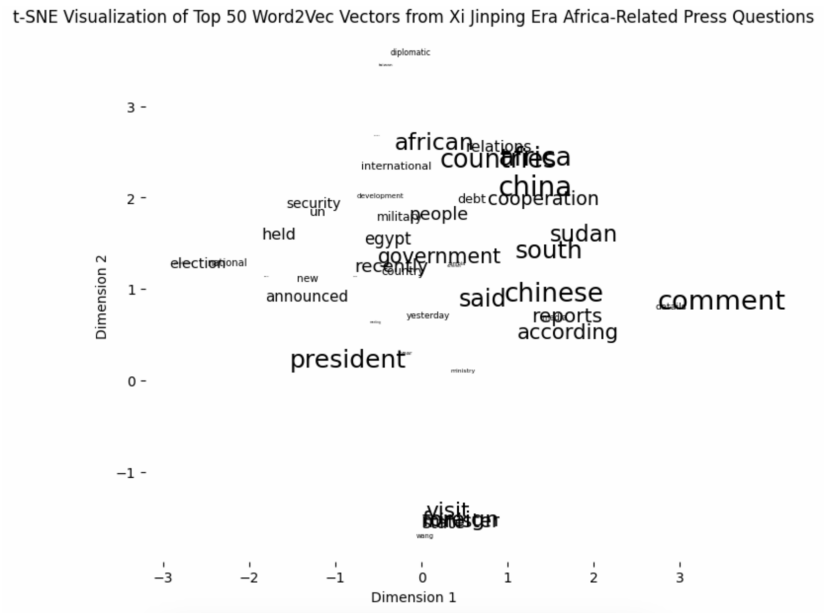


Figure 12. Visualization of Top 50 Word2Vec Vectors from the Xi Jinping Era Africa-Related Press Questions. Visualization on two dimensions using t-SNE reduction of the Word2Vec vectors. The larger the word, the more frequent it is in the corpus.

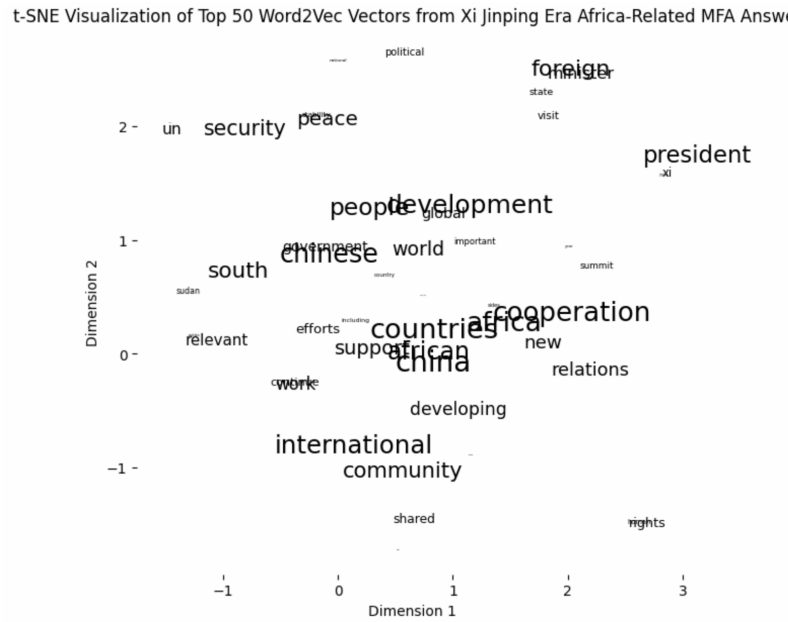


Figure 12. Visualization of Top 50 Word2Vec Vectors from the Xi Jinping Era Africa-Related MFA Answers. Visualization on two dimensions using t-SNE reduction of the Word2Vec vectors. The larger the word, the more frequent it is in the corpus.

Discussion

Africa-Related Question and Answer Patterns

The peaks in Africa-related questions and answers seen in 2007 and 2011 are easily explained by looking at the questions from this period. In 2007, Africa-related questions peaked in January, April, and May. Questions in January revolved around President Hu Jintao's visit to Africa, while those in April focused on the crisis in Sudan's Darfur and the PRC's involvement in the conflict. The May questions continue the focus on Sudan but add questions on the African Development Bank's meeting in Shanghai. As answers also peaked around this time, we can assume that they are related to these questions, and a similar assumption can be used to analyze the 2011 peak. In 2011, most of the Africa-related questions were in June and September, with

the June questions focusing on the crisis in Libya and the September questions following up on this topic, now with a focus on PRC's return to Libya.

This additional information underscores the fact that the press in 2007 and 2011, with Hu Jintao leading the PRC, wanted to know more about the PRC's engagement in conflicts that were highly visible in the West, and in which the West was concerned about the PRC's role. On the other hand, the PRC, which was slowly becoming more engaged on the world stage (as we saw in the background section earlier), was more than happy to signal its new shift by publishing English-language press conference transcripts that included these questions on the Sudan and Libya crises, as well as further answers about its other endeavors in Africa like the African Development Bank and Hu Jintao's visit to Africa. The fact that the PRC chose to showcase its shift and publicize it through the lens of Africa also underscores the importance of the region in these years to the PRC, fitting with the positive trend in PRC-Africa relations discussed at the start of this article.

However, once Xi Jinping came into power in 2013, the prevalence of Africa in press questions and MFA answers is much lower as both Africa-related questions and answers, as well as overall questions and answers, grow greatly beyond the Hu Jintao era numbers, almost doubling in some years. To be sure, the peaks in 2020 and 2021 for all questions and answers are clearly driven by the COVID-19 pandemic, which originated in 2020 in the PRC, but the consistent growth in questions and answers before these years and following Xi Jinping's ascent proves that such a trend is not solely due to the pandemic and the PRC's central role in it. In fact, this trend underscores the world's increasing overall interest in the PRC in the Xi Jinping years, as Xi makes the PRC presence in the world more felt through his flagship Belt and Road Initiative and other diplomatic, political, and economic connections. In turn, then, questions

about Africa and the PRC have become a smaller portion of the questions asked at the MFA press conference, as Xi moves from a tighter focus on Africa in the Hu Jintao era to global ambitions, a finding that opposes scholarly consensus on Xi's growing ambitions in Africa. These are, instead, growing at the same rate (and probably somewhat slower) when compared to his ambitions in other regions of the world.

Correspondence Analysis

It is no surprise to see the Hu Jintao and Xi Jinping years divided up quite well along the first dimension for both Africa-related questions and answers following correspondence analysis (Figures 8 and 9). What is intriguing is the second dimension, which, according to the screeplots for both datasets, is the cutoff for adding useful information to the analysis, though it only adds 14.29% for the questions dataset and 12.04% for the answers dataset. If we assume the first dimension to be Hu Jintao - Xi Jinping eras, with years to the left showcasing more of Hu Jintao's rhetoric and years to the right being those years that highlight Xi Jinping's rhetoric, what is the second dimension? Given that the extremes in this dimension for both datasets are represented by 2012 (only for the answers dataset) and 2013 on one end and 2009 on the other, the answer lies there. Looking at the counts for bigrams and trigrams for 2009 and 2013, the overall pattern seems to be that 2009 holds more PRC-focused words like Hu Jintao, foreign minister, President Hu Jintao, and does China, while 2013 holds more outside-focused words like current situation, South Sudan, Al Bashir, and President Omar.

Consequently, given that Hu Jintao started out more careful in the foreign policy and Africa sphere and gradually got more aggressive, as we saw in the background section, while Xi Jinping is focused on increasing the PRC's global presence, the first dimension for Africa-related questions and answers can be considered foreign policy aggressiveness in Africa from low (left)

to high (right), while the second dimension can be considered PRC-focused (high) or world-focused (low). Thus the location of the years tells one how focused the questions or answers that year were on foreign policy aggressiveness (dimension 1) or the PRC vs the globe (dimension 2). From this, and especially since questions and answers follow the same broad pattern, we can see that, beyond knowing that Xi's foreign policy rhetoric is stronger than Hu's, both eras vary in journalists' interest in the PRC itself vs bringing up global issues that the PRC is or might be involved with, as well as the PRC's interest in showcasing both these angles. In recent years, though, questions have turned to be more PRC-first and highly aggressive, and the PRC has wanted to showcase this method of focusing on Africa.

Word2Vec

With the Word2Vec analysis and the visualizations of the top 50 words across Hu Jintao and Xi Jinping eras, it is clear that Hu Jintao's PRC had journalists that were focused on the PRC's role in Sudan, Libya, Egypt, and Zimbabwe, with these words clustering together closely in the visualization of the questions, and, as discussed above with the descriptive analysis, Hu's PRC wanted to promote this focus to the international audience of its English-language press conference transcripts. On the other hand, MFA answers in this period focused on cooperation, relations, and countries (all clustered together and fairly large), as well as Sudan and Libya, but to a smaller degree, and the international community, its efforts, and the parties to conflicts (all also clustered together and fairly large). As a result, in the Hu era, the PRC was promoting, as discussed earlier, its engagement in African conflicts in which the West was also engaged, while also promoting an equal partnership with African countries and its increasing presence in the world community.

In the Xi Jinping era, questions focus instead more closely on China's relationship with African countries and their cooperation, given the close proximity of all these words in the top right corner of the visualization and their large size. Sudan does appear again in the context of South Sudan, but it is not as relevant as it was under Hu Jintao. The visualization of the answers similarly clusters China, countries, support, and Africa, as well as cooperation, echoing the main themes of the questions and adding international and community nearby. People and development also cluster together. In contrast to the Hu Jintao era, then, which was characterized by questions on specific African issues and answers attempting to promote the PRC's vision for its relationship with Africa, the Xi Jinping era is characterized by both press and state being on the same page about the PRC's vision for its relationship with Africa, with the MFA continuing to attempt to include the PRC in the international community and bring in new rhetoric on development. The PRC, then, through these press transcripts, continues to showcase the African relationship rhetoric that started under Hu Jintao while trying to place itself in the international community more forcefully.

Conclusion

With the goal of showcasing the power of quantitative analysis for PRC research, this article took an innovative corpus of the PRC's Ministry of Foreign Affairs daily press conferences and applied three diverse quantitative methods to analyze PRC-Africa relationship rhetoric under Hu Jintao and Xi Jinping—basic counts and proportions, correspondence analysis with bigrams and trigrams by year, and Word2Vec and t-SNE for visualization. These methods substantiated one of the main rhetoric trends discussed in the qualitative literature, the increase in foreign policy aggressiveness from the Hu Jintao to Xi Jinping eras, but they also contested the belief that Xi Jinping is becoming more aggressive on Africa specifically, instead pointing out

that he is becoming more aggressive across the world. Furthermore, these methods uncovered two other aspects of PRC-Africa relationship rhetoric: an additional dimension beyond foreign policy aggression to divide years within and across the Hu Jintao and Xi Jinping eras (China-focused vs. world-focused) and the gradual shift of partnership rhetoric from being purely from the spokesperson (state) side during the Hu Jintao era to being part of journalists' vocabulary in the Xi Jinping era, a change the PRC pushed and and the main Africa rhetoric it wants to keep impressing on the world.

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